Assignment: Object Detection by YOLO

Fine-tune a pre-trained object detection model, such as **YOLO**, to detect and classify custom real-world objects captured using mobile devices or cameras that are not adequately recognized by the original pre-trained model.

To complete the assignment, perform the following tasks:

1. Custom Dataset Collection
    - Select one or more favourite real-world object categories that you would like the model to detect.
    - Capture images of your selected objects using your smartphone or digital camera.
    - Collect a sufficiently diverse set of images under different real-world conditions, including:
        - Varying illumination
        - Different and complex backgrounds
        - Different viewing angles
        - Partial object occlusion
        - Different object scales and distances
        - Different object orientations and positions
    - Ensure that the collected images represent realistic conditions in which the objects may be encountered.
2. Dataset Annotation and Preparation
    - Annotate the target objects using appropriate bounding-box annotations.
    - Assign the correct class label to each annotated object.
    - Divide the dataset into training, validation, and testing sets using an appropriate split.
3. Data Augmentation
    - Apply suitable data augmentation techniques to increase the diversity of the training data and improve the robustness and generalization capability of the detector.
    - Examples may include image flipping, rotation, scaling, cropping, translation, and changes in brightness or contrast.
4. Model Fine-Tuning
    - Select a suitable pre-trained object detection model, such as YOLO.
    - Fine-tune the model using the custom training dataset.
    - Document the model architecture, training configuration, hyperparameters, and number of training epochs.
5. Performance Evaluation

    - Evaluate the fine-tuned model on the test set using standard object detection metrics, including:
        - Precision
        - Recall
        - Mean Average Precision (mAP)
        - Inference speed
        - Detection latency
6. Comparison with the Original Pre-trained Model
    - Evaluate the original pre-trained model on the same custom test dataset.
    - Compare its performance with that of the fine-tuned model for the selected custom object categories.
    - Discuss the changes in detection performance after fine-tuning.
7. Analysis
    - Analyze the strengths and limitations of the developed detector.
    - Identify challenging cases where the model produces false positives, false negatives, or inaccurate bounding boxes.
    - Discuss how factors such as illumination, background complexity, occlusion, viewing angle, and object scale affect detection performance.

Report Requirements

The report should include the following sections:

1. Introduction and Objective
2. Dataset Collection, Preparation, and Annotation Methodology
3. Data Augmentation Strategies
4. Model Architecture and Training Configuration
5. Experimental Setup
6. Quantitative Results
    - Precision
    - Recall
    - mAP
    - Inference speed
    - Detection latency
7. Comparison Between the Original Pre-trained and Fine-tuned Models
8. Qualitative Detection Results
9. Performance Analysis
10. Strengths and Limitations
11. Discussion
12. Conclusion

Expected Outcome

The assignment should demonstrate how **transfer learning and fine-tuning can adapt a pre-trained object detection model to recognize personally selected custom objects in real-world images**, particularly objects that are not adequately detected by the original pre-trained model.